Exercise 1

TID	Items
T1	F, G, I, J
T2	G, H, J
T3	G, I, J
T4	G, H, J
T5	G, H, I, J
T6	G, I, J

A database contains **6 transactions**. Let:

- **Minimum support = 50%**
- **Minimum confidence = 75%**

Tasks:

1. Find **all frequent itemsets**. Show every step of your calculation.
   For each frequent itemset, indicate its **support**.
2. Give **3 examples of strong association rules** discovered from the database.
   For each rule, indicate its:
   - **support**
   - **confidence**
   - **lift**

Submission: Student_ID_HW#8_Exercise-1.pdf containing your answer

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Exercise 2

Repeat Exercise 1 using **RapidMiner** or **Python**.

Tasks:

1. Find **all frequent itemsets** and indicate the **support** of each itemset. **Include screenshots**.
2. Give **3 examples of strong association rules** discovered from the database.
   For each rule, indicate its:
   - **support**
   - **confidence**
3. **Include screenshots**.

Submission:

- Student_ID_HW#8_Exercise-2.pdf containing your answer
- Student_ID_HW#8_Exercise-2.xml or Student_ID_HW#8_Exercise-2.ipynb

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Exercise 3 Case-Study: Customer Profiling for PEP Product using Association Rules

A financial institution recently launched a new investment product called **PEP (Personal Equity Plan)**. The marketing department sent promotional letters to existing customers and recorded whether customers purchased the product. The goal of this analysis is to **discover customer profiles that are likely to purchase PEP**, using **Association Rule Mining**.
Dataset: **bank-data.csv**

Attributes include:

- Demographic attributes (sex, age, region, married)
- Financial attributes (income, car ownership, mortgage, savings account, current account)
- Target variable: **pep (YES/NO)**

Part A — Data Preparation (Preprocessing)

A1. Attribute Selection

From the dataset, select **6–10 attributes** that you believe are most relevant for discovering customer behavior.
Explain briefly:

- Why you selected these attributes
- Why you removed some attributes

A2. Convert Categorical Attributes

Convert categorical attributes into **binary variables (one-hot encoding)**.
Example:

Original	Transformed
sex = MALE	sex_MALE = 1
sex = FEMALE	sex_FEMALE = 1

Show:

- Screenshot of transformation
- Or table showing before / after

A3. Discretization of Numerical Attributes

Convert **numerical attributes** into categorical groups.
Example:

Attribute	Possible Groups

age	young, middle, senior
income	low, medium, high
children	0, 1–2, 3+

Explain:

- The discretization method used
- Why the grouping makes sense

A4. Final Transaction Format

Prepare the final dataset in **transaction format** suitable for association rule mining.
Example transaction:

```
{sex_FEMALE, married_YES, income_HIGH, mortgage_NO, pep_YES}
```

Show **5 example transactions** from your dataset.

Part B — Association Rule Mining

B1. Parameter Selection

Choose values for:

- **Minimum Support**
- **Minimum Confidence**

Explain your reasoning:

- Why the values are appropriate
- What happens if they are too high or too low

B2. Frequent Itemsets

Show the **top frequent itemsets discovered**.
Present results in a table:

Itemset	Support
{income_HIGH, married_YES}	0.xx
{income_HIGH, pep_YES}	0.xx

Include a **screenshot of the output from the tool**.

B3. Association Rules

Show at least **5 association rules** with:

Rule	Support	Confidence	Lift
income_HIGH → pep_YES	0.xx	0.xx	1.xx

Highlight rules where:

- **confidence ≥ threshold**
- **lift > 1**

Include a **screenshot of the results**.

B4. Interpretation of the 2 Most Interesting Rules

Select **two rules that you find most interesting**.
For each rule explain:

1. What the rule means in plain language
2. What type of customer profile it represents

Example:
Rule:

```
income_HIGH & married_YES → pep_YES
```

Interpretation:
Customers who are **married and have high income** are more likely to purchase the PEP product.
This may occur because these customers have **higher financial stability and investment capacity**.

Submit:
1️⃣ **Report (PDF or Word)** including

- preprocessing explanation
- screenshots
- rules discovered
- interpretation

2️⃣ Workflow / Code file

- RapidMiner → .xml
- Python → .ipynb

File names:

```
Student_ID_HW#8_Exercise-3.pdf
Student_ID_HW#8_Exercise-3.xml or
Student_ID_HW#8_Exercise-3.ipynb
```